Section A [30 marks]

Answer all the question in the spaces provided on the question paper.

If working is needed for any question, it must be shown in the space below that question.

1. **(a) Figure 1** shows one of the driving tools used in the Design and Technology workshop.

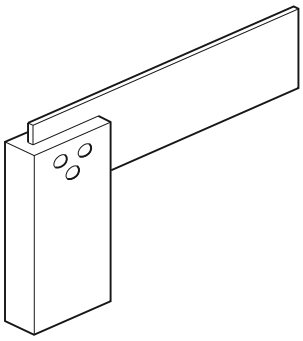


Figure 1

(i) Give the name of the tool in **figure 1**.

..... [1]

(ii) State **two** safety rules to be observed when using the tool in **figure 1**.

1.....

.....

2.....

..... [2]

(iii) Give **two** uses of the tool in **figure 1** above.

1.....

2..... [2]

(b) State **one** property of ash which makes it suitable for the mallet handle.

…… [1]

[Total: 6 marks]

2. **Figure 2** shows one of the tools used in the Design and Technology workshop.

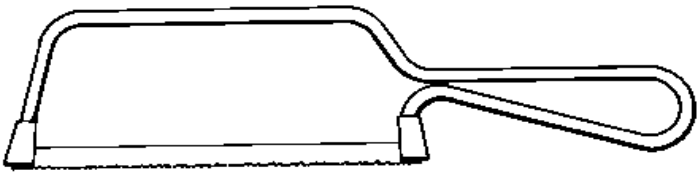


Figure 2

(a) i. What is the name of the tool?

…… [1]

ii. What is the use of the tool in **figure 2** above?

……

…… [1]

(b) Pulleys and gears are both used in the transmission of motion.

(i) Give **one** advantage of gears over pulleys.

.....

..... [2]

(ii) Mention **one** example where gears are used.

.....

..... [2]

[Total: 6 marks]

3. **Figure 3** shows a complete joint used on boxes and cabinet drawers.

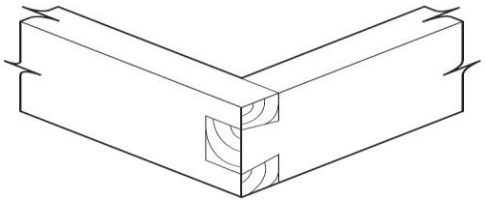


Figure 3

(i) Name the joint [1]

(ii) Name and sketch a saw used for making the joint above [4]

[Total: 6 marks]

4. Complete the table below which shows electrical energy being converted Into various forms.

Type of Energy	Component	Converted into:
Electrical Energy	Bulb	
Electrical Energy	Loud speaker	
Electrical Energy	Motor	

[Total: 6 marks]

5. Figure 4 below shows some of the materials used in design and technology workshop. Study it and fill in the blanks.

MATERIAL	PROPERTIES	USE
Brass		
Bronze		
Aluminium	Light in weight	Making aircraft bodies

(b) List two fixing devices that can be used for security and movement purposes.

i. Movement: …… [1]

ii. Security: …… [1]

[Total: 6 marks]

Section B [80 marks]

There are three parts in this section. Part I, is compulsory. Part II and III, answer one question from each part.

Part I Compulsory.

6. Study **Figure 4** and refer to it as you answer the questions that follow.

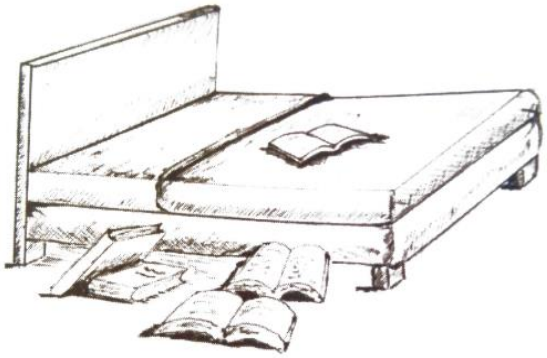


Figure 4

(a) (i) Write down the situation. [2]

(ii) Identify the problem from the situation in **figure 4**. [2]

(b) (i) Derive a design brief from the problem in **(a) (ii)** above. [2]

(ii) Using annotations, sketch one possible solution in pictorial view for the situation above. [12]

(c) List **three** specifications which could be considered when coming up with your design solution. [3]

(d) You are also expected to do research in order to gather information that will guide you in making decisions. Mention **four** possible areas of research. [4]

[Total 25 marks]

Part II Answer one question

7. a. Figure 5 shows two filing techniques used shaping materials.



A **B**

Figure 5

(i) Name the two filing techniques and briefly describe how each one is done. [4]

(ii) What is a business idea? [1]

(iii) List the business ideas in Design and Technology [3]

(iv) What is the difference between mass production and butch production? [2]

(v) What are aggregates? List three fine aggregates and two coarse aggregates [7]

b. A designer has to have knowledge of the many different methods of joining materials.

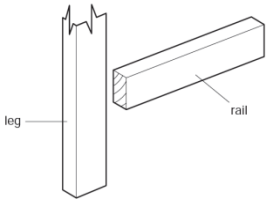
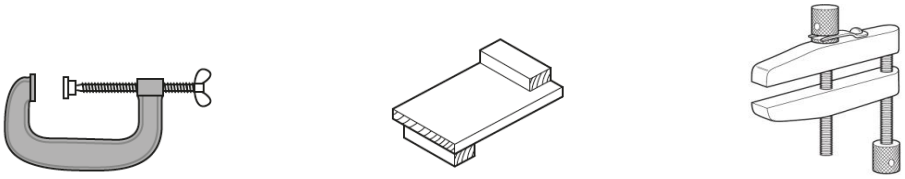


Figure 6

(i) Using notes and sketches describe how you would join a wooden chair rail to a wooden leg with a mortise and tenon joint. [5]

(ii) Figure 7 shows holding tools used in the workshop.



A B C

Figure 7

(a) Name the tools A, B and C. [3]

[Total: 25 marks

8. **Figure 8** shows a work piece being drilled. The part marked **'X'** is slot for a key hole.

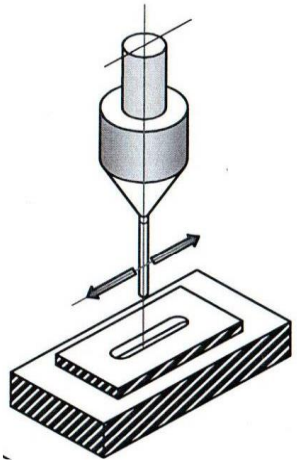


Figure 8

(a) List **two** reasons why it is necessary to place scrap wood underneath the work piece when drilling. [2]

(b) **(i)** Illustrate, with the aid of sketches, how you would obtain the slot „**X**". [12]

(ii) Give examples of **two** materials you can make a slot in. [2]

(c) **(i)** Classify **three** sources of energy. [3]

(ii) Using sketches demonstrate the following motions:

1. Reciprocating [2]
2. Oscillating [2]
3. Rotary [2]

[Total: 25 marks]

9. **Figure 9** shows an incomplete design for a reading lamp to be made in a Design and Technology Workshop. The arms, base and supports are made from hardwood.

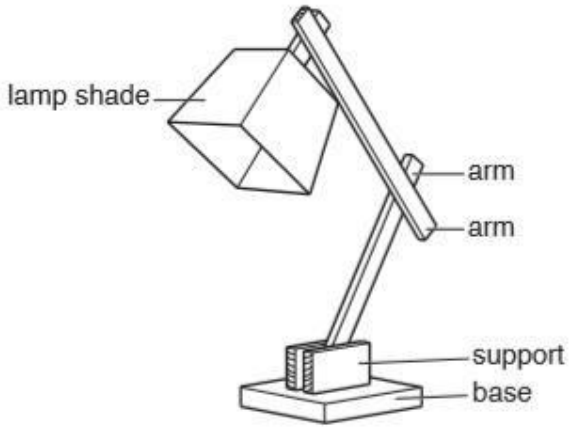
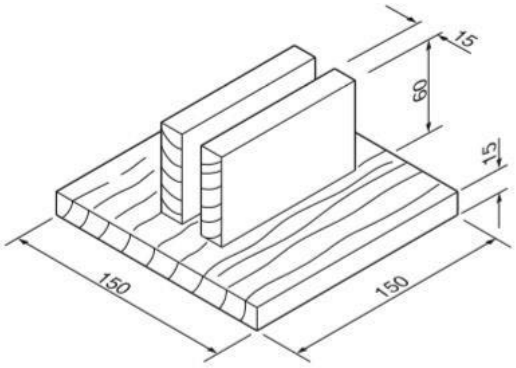


Figure 9

(a) The figure below shows details of the base and support,



Use sketches and notes to show how the base and support could be joined permanently. [4]

b. Complete the drawing below to show an exploded view of a stopped housing joint.

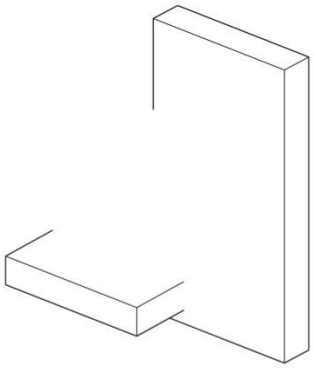


Figure 10 [4]

c. Figure 11 below shows one of the types of bonds in bricklaying.

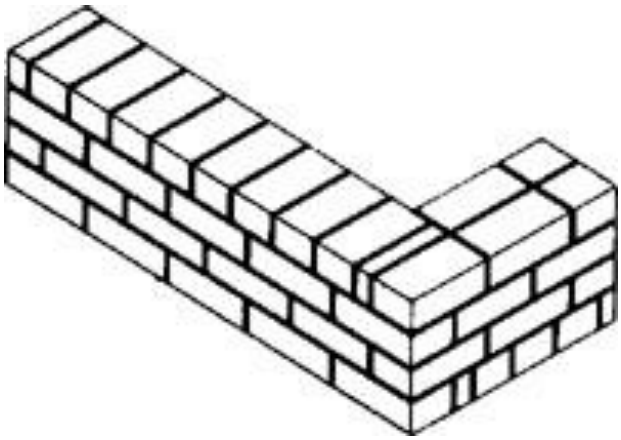


Figure 11

a. Name the bond [1]

b. State **two** purpose of bonding in bricklaying [2]

c. There are a variety of tools used when jointing and pointing. Differentiate between Jointing and pointing. [2]

d.Give **two** reasons for jointing and pointing in bricklaying. [2]

e. Mention any **two** types of tools used during the process of jointing and pointing. [2]

f. **Table 3** below shows electrical symbols. Name each symbol.

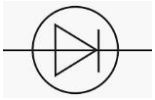
	SYMBOL
A.	
B.	

Table 3

[2]

(a)

i. Name symbol A

ii. Name symbol B

(b) Electronic components are classified as input or output. Classify the following components.

(i) Motor

(ii) Diode

(iii) Loud speaker

(iv) Transistor

(v) Microphone

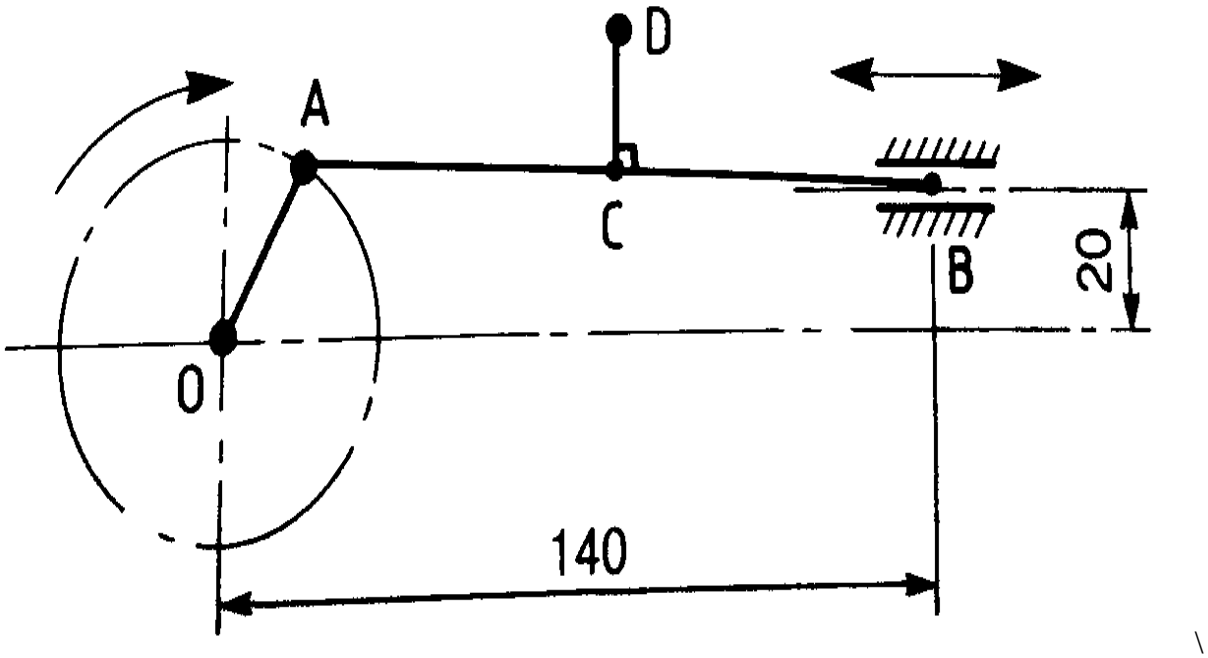
(vi) LDR

[6]

[Total: 25 marks]

PART III Answer one question

10. The diagram below shows the link mechanism. Crank **AO** rotates about **O** in a clockwise direction. The end **B** of link **AB** is restricted to move in a horizontal straight line as indicated. **C** is the mid-point of **AB** and dc is perpendicular to **AB**. Plot a locus of point d if **OA** = 30mm, **AB** = 130 mm and **DC** 20 mm.



\

[Total: 30 marks]

10. Figure 12 shows a proposed design of an aero plane logo for Zambia Airways Corporation drawn by Design and Technology learners at Katete Girls' Secondary School.

The curve **ABC** is a parabola with the vertex at **B.**

The curve **AED** is elliptical with focal points at **F** and **F1.**

The line **CD** is a tangent to the ellipse.

The five rectangular windows are the same size and are equally spaced.

Draw the given design.

Show all construction clearly.

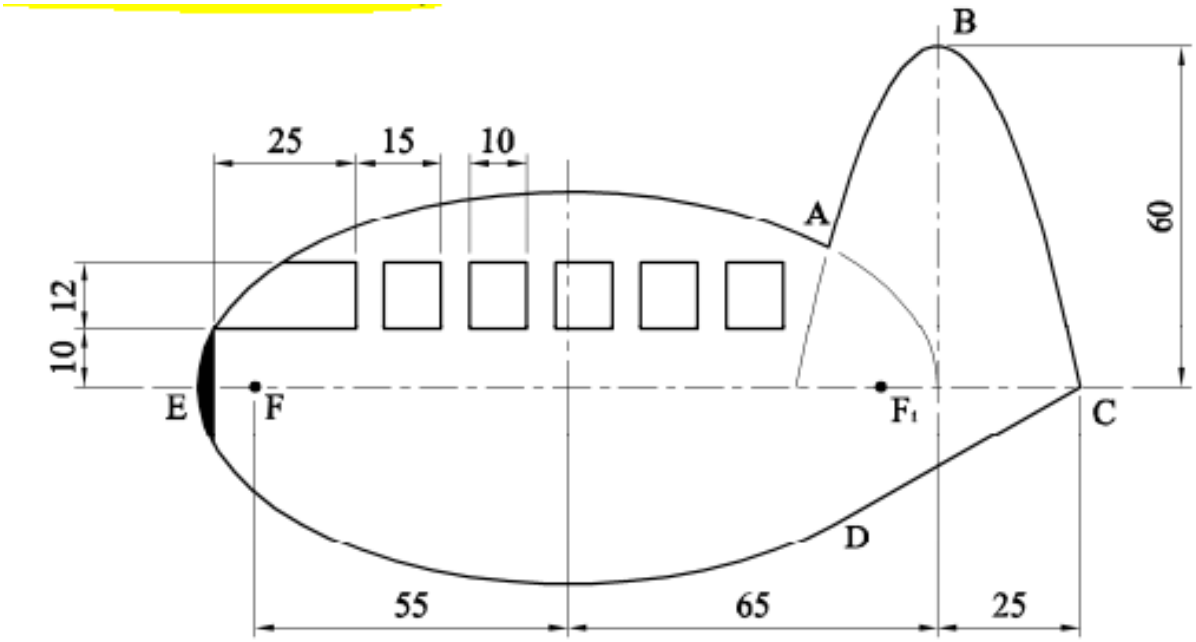


Figure 12

[Total: 30 marks]

END OF EXAM…..